



Moove Education Platform

Software Requirements Specification (SRS)

Version 3.0

July 19, 2026

Prepared By:

Moove

This product includes software from the Open edX project, licensed under Apache 2.0.

July 19, 2026

Contents

0.1	Introduction	3
0.1.1	Purpose	3
0.1.2	Scope	3
0.1.3	Actors	3
0.2	Functional Requirements – Educator Features	3
0.2.1	Course Creation & Planning Module	3
0.2.2	Instructional Design Concepts	5
0.2.3	Learner Engagement & Communication Module	5
0.2.4	Content Creation & Management Module	6
0.2.5	Components & Activities Module	7
0.2.6	Data & Analytics Module	7
0.2.7	Accessibility Module	8
0.2.8	Import/Export / OLX Module	9
0.3	Functional Requirements – Learner Features	9
0.3.1	Dashboard & Profile Module	9
0.3.2	Course Navigation Module	11
0.3.3	Enrollment Module	11
0.3.4	Course Prerequisites Module	11
0.3.5	Course Content Access Module	12
0.3.6	Course Progress Module	13
0.3.7	Certificates & Badges Module	13
0.3.8	Video Player Module	14
0.3.9	Discussions Module	14
0.3.10	Assignments & Assessments Module	15
0.3.11	Team Projects Module	16
0.3.12	Search Module	17
0.3.13	Bookmarking Module	18
0.3.14	Notifications & Preferences Module	18
0.3.15	Course Wiki Module	19
0.4	Security and Compliance Requirements (Non-Functional)	19
0.4.1	Access Control (CMCSRS 6.17)	19
0.4.2	Cryptography (CMCSRS 6.18)	20
0.4.3	Network Security (CMCSRS 5.7.c)	20
0.4.4	Logging and Monitoring (CMCSRS 6.25)	20
0.4.5	Incident Response and Business Continuity (CMCSRS 6.26, 6.19)	20
0.4.6	Personnel Security (CMCSRS 5.6, 6.37)	21
0.4.7	Compliance and Audit (CMCSRS 6.21, 6.22, 6.23)	21
0.5	Traceability Matrix	21
0.6	References	22

.1	Full Apache License 2.0 Text	23
----	--	----

0.1 Introduction

0.1.1 Purpose

This document defines the functional and non-functional requirements for the Moove Education Platform, an Open edX-based learning management system deployed using Tutor. It serves as the living source of truth for system capabilities, guiding continuous development, internal quality assurance, and security assessment activities. The requirements are aligned with Ethiopian regulatory frameworks, including the Critical Mass Cyber Security Requirement Standard (CMCSRS) and the National Cybersecurity Policy.

0.1.2 Scope

The system provides a comprehensive online learning environment supporting:

- Course creation, authoring, and management (Educator features).
- Learner enrollment, progression, and engagement (Learner features).
- Content delivery (video, text, interactive components).
- Assessments (quizzes, assignments, peer reviews, proctored exams).
- Collaboration (discussions, teams, wikis).
- Analytics and reporting.
- Administrative controls.

0.1.3 Actors

- **Educator (Instructor/Staff):** Creates and manages courses, creates content, assesses learners, moderates discussions, views analytics.
- **Learner (Student):** Enrolls in courses, accesses content, completes assessments, participates in discussions, tracks progress.
- **Administrator:** Manages users, courses, site configuration, and system health.
- **System:** Automated processes (notifications, email, background tasks).

0.2 Functional Requirements – Educator Features

This section defines requirements for instructors and course staff.

0.2.1 Course Creation & Planning Module

General Description Provides tools for educators to create, plan, and structure courses.

Requirement UFR-OE-001-01: Create a New Course

1. **Title:** Create a new course.
2. **UML Traceability:** Use Case: "Create Course"; Activity: "Course Creation Flow".
3. **Actor:** Educator (Instructor/Staff).
4. **Pre-conditions:** Educator is logged in with course creation permissions.
5. **Post-conditions:** Course is created with basic metadata; course skeleton is initialized.
6. **Normal Flow:** 1. Educator navigates to Studio (CMS). 2. Clicks "New Course". 3. Enters course title, organization code, course number, and course run. 4. Sets start date, enrollment start/end dates, and course visibility. 5. Clicks "Create". 6. System initializes course structure and creates course ID.
7. **Alt Flows:** Invalid course number/org code → validation error.
8. **Business Rules:** Course ID must be unique. Course number and organization code follow defined patterns.
9. **Input/Output:** Input: title, org, number, run, start date, enrollment dates. Output: Course object with ID.
10. **Acceptance:**

Given I am an educator
When I create a course with title "Introduction to Physics"
Then I receive a 201 Created
And the course appears in my course list

Requirement UFR-OE-001-02: Plan a New Course (Outline)

1. **Title:** Plan course outline.
2. **Actor:** Educator.
3. **Pre:** Course exists.
4. **Post:** Course outline (sections, subsections, units) is defined.
5. **Normal Flow:** Educator adds sections, subsections, and units to organize content.
6. **Input/Output:** Input: section/subsection/unit metadata. Output: updated course structure.

Requirement UFR-OE-001-03: Enroll Learners in a Course

1. **Title:** Enroll learners manually.
2. **Actor:** Educator.
3. **Pre:** Course exists and enrollment is open.
4. **Post:** Learner is enrolled.
5. **Normal Flow:** Educator enters learner email(s) and assigns enrollment track (audit/verified).
6. **Alt:** Bulk enrollment via CSV upload.
7. **Acceptance:** Bulk upload processes 500+ enrollments.

0.2.2 Instructional Design Concepts

Requirement UFR-OE-002-01: Apply Instructional Design Principles

1. **Title:** Apply instructional design concepts.
2. **Actor:** Educator.
3. **Normal Flow:** Educator structures course using learning objectives, assessments, and activities following Open edX best practices.
4. **Business Rules:** Supports backward design, Bloom's taxonomy alignment.

0.2.3 Learner Engagement & Communication Module

Requirement UFR-OE-003-01: Course Discussions

1. **Title:** Configure course discussions.
2. **Actor:** Educator.
3. **Pre:** Course exists.
4. **Post:** Discussion forums enabled.
5. **Normal Flow:** Educator enables discussions, configures topics, and sets moderation policies.

Requirement UFR-OE-003-02: Bulk Emails

1. **Title:** Send bulk emails to learners.
2. **Actor:** Educator.
3. **Pre:** Course has enrolled learners.
4. **Normal Flow:** Educator composes email, selects recipients (all learners, specific segments), sends.
5. **Business Rules:** Email is sent asynchronously via Celery.

Requirement UFR-OE-003-03: Course Updates

1. **Title:** Publish course updates.
2. **Actor:** Educator.
3. **Normal Flow:** Educator creates update post (announcement) visible on course home page.

0.2.4 Content Creation & Management Module

Requirement UFR-OE-004-01: Create and Manage Components

1. **Title:** Add and manage course components.
2. **Actor:** Educator.
3. **Pre:** Course structure exists.
4. **Post:** Component added to unit.
5. **Component Types:** HTML, Video, Problem, Discussion, Poll, Survey, and custom XBlocks.

Requirement UFR-OE-004-02: Manage Text Components

1. **Title:** Create and edit HTML/text components.
2. **Actor:** Educator.
3. **Normal Flow:** Educator uses rich text editor or raw HTML to create text content.
4. **Features:** Supports LaTeX, Code snippets, Embedded media.

Requirement UFR-OE-004-03: Manage Video Components

1. **Title:** Upload and manage video content.
2. **Actor:** Educator.
3. **Pre:** Video file available.
4. **Normal Flow:** Educator uploads video file or enters YouTube/Vimeo URL; system processes and displays video player.
5. **Features:** Transcript support, speed control, download options.

Requirement UFR-OE-004-04: Control Content Visibility & Access

1. **Title:** Control content visibility and access.
2. **Actor:** Educator.
3. **Normal Flow:** Educator sets release dates, prerequisites, and visibility rules for content.
4. **Features:** Release date, prerequisite, visibility to specific user groups.

0.2.5 Components & Activities Module

Requirement UFR-OE-005-01: Dropdown Problem

1. **Title:** Create dropdown problem.
2. **Actor:** Educator.
3. **Description:** Multiple choice with a dropdown selector; learner selects one option.

Requirement UFR-OE-005-02: Multi-Select Problem

1. **Title:** Create multi-select problem.
2. **Description:** Learner selects multiple correct answers from a list.

Requirement UFR-OE-005-03: Open Response Assessments (ORA)

1. **Title:** Create open response assessment.
2. **Actor:** Educator.
3. **Normal Flow:** Educator defines prompt, rubric, and assessment steps (self, peer, staff).
4. **Features:** Rubric-based grading, peer review, self-assessment.

Requirement UFR-OE-005-04: Single Select Problem

1. **Title:** Create single select (radio button) problem.
2. **Description:** Learner selects one correct answer from a list of options.

Requirement UFR-OE-005-05: Text Input Problem

1. **Title:** Create text input problem.
2. **Description:** Learner enters a short text answer; system checks against expected answer(s).

0.2.6 Data & Analytics Module

Requirement UFR-OE-006-01: View Course Enrollments

1. **Title:** View course enrollment data.
2. **Actor:** Educator.
3. **Normal Flow:** Educator accesses enrollment dashboard showing totals, trends, and demographics.

Requirement UFR-OE-006-02: View Learner Data

1. **Title:** View learner data and activity.
2. **Actor:** Educator.
3. **Normal Flow:** Educator views individual learner progress, engagement metrics, and activity history.

Requirement UFR-OE-006-03: View Learner Grades

1. **Title:** View and manage learner grades.
2. **Actor:** Educator.
3. **Normal Flow:** Educator accesses gradebook, views grades by assignment and learner.
4. **Features:** Export to CSV, manual grade adjustment.

Requirement UFR-OE-006-04: View Answer Data

1. **Title:** View answer distribution data.
2. **Actor:** Educator.
3. **Normal Flow:** Educator views analytics on problem answers (e.g., % correct per problem).

Requirement UFR-OE-006-05: View Certificate Data

1. **Title:** View certificate issuance data.
2. **Actor:** Educator.
3. **Normal Flow:** Educator views list of issued certificates and download statistics.

0.2.7 Accessibility Module

Requirement UFR-OE-007-01: Open edX Accessibility Guidelines

1. **Title:** Ensure accessibility compliance.
2. **Actor:** Educator.
3. **Description:** Course content must follow Open edX accessibility guidelines (WCAG 2.1 AA).

Requirement UFR-OE-007-02: Mobile Experience Design

1. **Title:** Design for mobile experience.
2. **Actor:** Educator.
3. **Description:** Course content must be responsive and functional on mobile devices.

Requirement UFR-OE-007-03: Accessibility Best Practices Checklist

1. **Title:** Follow accessibility best practices.
2. **Actor:** Educator.
3. **Description:** Use alt text, proper heading structure, accessible color contrast, keyboard navigation.

Requirement UFR-OE-007-04: Manage Video Transcripts

1. **Title:** Upload and manage video transcripts.
2. **Actor:** Educator.
3. **Normal Flow:** Educator uploads transcript files (.srt, .vtt) for video components.

0.2.8 Import/Export / OLX Module

Requirement UFR-OE-008-01: Open Learning XML (OLX) Support

1. **Title:** Support for Open Learning XML.
2. **Actor:** Educator.
3. **Description:** Course content can be exported and imported using OLX format.

Requirement UFR-OE-008-02: Getting Started with OLX

1. **Title:** Access OLX documentation and tools.
2. **Actor:** Educator.
3. **Description:** Provide resources for educators to learn and use OLX.

0.3 Functional Requirements – Learner Features

This section defines requirements for learners.

0.3.1 Dashboard & Profile Module

Requirement UFR-OE-101-01: Explore Dashboard and Profile

1. **Title:** Access and explore learner dashboard.
2. **Actor:** Learner.
3. **Pre:** Learner is logged in.
4. **Post:** Dashboard displays enrolled courses, progress, and announcements.

Requirement UFR-OE-101-02: Access Courses from Dashboard

1. **Title:** Access enrolled courses from dashboard.
2. **Actor:** Learner.
3. **Normal Flow:** Learner clicks on course card to enter the course.

Requirement UFR-OE-101-03: Share Courses on Social Media

1. **Title:** Share course links on social media.
2. **Actor:** Learner.
3. **Normal Flow:** Learner clicks "Share" button, selects platform (Facebook, Twitter, LinkedIn).

Requirement UFR-OE-101-04: Update Profile Image

1. **Title:** Upload or update profile image.
2. **Actor:** Learner.
3. **Normal Flow:** Learner uploads image file; system resizes and stores.

Requirement UFR-OE-101-05: Update Profile Details

1. **Title:** Update profile information.
2. **Actor:** Learner.
3. **Fields:** Name, email, location, bio, language preferences.

Requirement UFR-OE-101-06: Add Social Media Links

1. **Title:** Add personal social media links.
2. **Actor:** Learner.
3. **Description:** Learner adds links to Twitter, LinkedIn, GitHub, etc.

Requirement UFR-OE-101-07: Toggle Profile Visibility

1. **Title:** Control profile visibility.
2. **Actor:** Learner.
3. **Normal Flow:** Learner toggles visibility of profile sections (public/private).

Requirement UFR-OE-101-08: View Another Learner's Profile

1. **Title:** View other learners' profiles.
2. **Actor:** Learner.
3. **Normal Flow:** Learner clicks on username in discussions or course roster.

Requirement UFR-OE-101-09: Update Course-Specific Settings

1. **Title:** Update course-specific settings.
2. **Actor:** Learner.
3. **Normal Flow:** Learner configures per-course preferences (e.g., email notifications, language).

0.3.2 Course Navigation Module

Requirement UFR-OE-102-01: View Course Sections from Navigation Sidebar

1. **Title:** Navigate course using sidebar.
2. **Actor:** Learner.
3. **Normal Flow:** Sidebar displays course sections and subsections for easy navigation.

0.3.3 Enrollment Module

Requirement UFR-OE-103-01: Enroll in a Course

1. **Title:** Enroll in a course.
2. **Actor:** Learner.
3. **Pre:** Course is open for enrollment.
4. **Post:** Learner is enrolled.
5. **Normal Flow:** Learner clicks "Enroll" on course page; system confirms enrollment.

Requirement UFR-OE-103-02: Open and Start a Course

1. **Title:** Open and start a course.
2. **Actor:** Learner.
3. **Post:** Learner accesses course content.

0.3.4 Course Prerequisites Module

Requirement UFR-OE-104-01: Overview of Prerequisites

1. **Title:** View course prerequisites.
2. **Actor:** Learner.
3. **Normal Flow:** Learner views prerequisite requirements before enrolling.

Requirement UFR-OE-104-02: Complete Prerequisite Course

1. **Title:** Complete prerequisite course.
2. **Actor:** Learner.
3. **Normal Flow:** Learner must complete prerequisite course before enrolling in advanced course.

Requirement UFR-OE-104-03: Pass Entrance Exam

1. **Title:** Pass entrance exam.
2. **Actor:** Learner.
3. **Normal Flow:** Learner takes and passes an entrance exam to gain access.

0.3.5 Course Content Access Module**Requirement UFR-OE-105-01: Troubleshoot Accessing Course Content**

1. **Title:** Troubleshoot content access issues.
2. **Actor:** Learner.
3. **Normal Flow:** System provides error messages and support for content access issues.

Requirement UFR-OE-105-02: Access an Archived Course

1. **Title:** Access archived course.
2. **Actor:** Learner.
3. **Normal Flow:** Learner can view read-only archived course content.

Requirement UFR-OE-105-03: View Final Course Grade

1. **Title:** View final course grade.
2. **Actor:** Learner.
3. **Post:** Learner sees final grade and passing status.

Requirement UFR-OE-105-04: View Certificate Status

1. **Title:** View certificate status.
2. **Actor:** Learner.
3. **Normal Flow:** Learner views certificate issuance status and download link.

0.3.6 Course Progress Module

Requirement UFR-OE-106-01: Find Course Start Date

1. **Title:** View course start date.
2. **Actor:** Learner.
3. **Normal Flow:** Course start date displayed on course info and dashboard.

Requirement UFR-OE-106-02: Start or Resume a Course

1. **Title:** Start or resume a course.
2. **Actor:** Learner.
3. **Normal Flow:** Learner clicks "Start" or "Resume" button; system navigates to last visited content.

Requirement UFR-OE-106-03: Progress Indicators

1. **Title:** View progress indicators.
2. **Actor:** Learner.
3. **Description:** Visual indicators show completion percentage for sections, subsections, and course overall.

Requirement UFR-OE-106-04: The Progress Page

1. **Title:** Access the Progress page.
2. **Actor:** Learner.
3. **Normal Flow:** Learner navigates to Progress page to view detailed completion status.

0.3.7 Certificates & Badges Module

Requirement UFR-OE-107-01: Accessing a Certificate

1. **Title:** Access earned certificate.
2. **Actor:** Learner.
3. **Post:** Learner downloads or prints certificate.

Requirement UFR-OE-107-02: Viewing Earned Badges in Profile

1. **Title:** View earned badges in profile.
2. **Actor:** Learner.
3. **Normal Flow:** Badges appear in profile; click to view details.

Requirement UFR-OE-107-03: Sharing Badges on Mozilla Backpack

1. **Title:** Share badges to Mozilla Backpack.
2. **Actor:** Learner.
3. **Normal Flow:** Learner pushes badge to Mozilla Backpack for portability.

0.3.8 Video Player Module**Requirement UFR-OE-108-01: Watching Videos on the Open edX Video Player**

1. **Title:** Use the video player.
2. **Actor:** Learner.
3. **Features:** Play, pause, volume, speed control, full-screen, CC/subtitles.

0.3.9 Discussions Module**Requirement UFR-OE-109-01: Getting Started with Discussions**

1. **Title:** Access course discussions.
2. **Actor:** Learner.
3. **Normal Flow:** Learner accesses discussions from course navigation.

Requirement UFR-OE-109-02: Finding and Following Conversations

1. **Title:** Find and follow conversations.
2. **Actor:** Learner.
3. **Features:** Search, filter, follow threads, get notifications.

Requirement UFR-OE-109-03: Taking Part in Discussions

1. **Title:** Participate in discussions.
2. **Actor:** Learner.
3. **Normal Flow:** Learner creates new threads, replies to existing ones.

Requirement UFR-OE-109-04: Staying Updated with Notifications

1. **Title:** Receive discussion notifications.
2. **Actor:** Learner.
3. **Normal Flow:** System sends email notifications for followed threads.

0.3.10 Assignments & Assessments Module

Requirement UFR-OE-110-01: Completing Mathematical and Scientific Assignments

1. **Title:** Complete mathematical assignments.
2. **Actor:** Learner.
3. **Features:** Support for LaTeX, math symbols, scientific notation.

Requirement UFR-OE-110-02: Taking a Timed Exam

1. **Title:** Take timed exam.
2. **Actor:** Learner.
3. **Normal Flow:** Learner starts exam; timer starts; auto-submit at expiration.

Requirement UFR-OE-110-03: Proctored Exams

1. **Title:** Take proctored exam.
2. **Actor:** Learner.
3. **Features:** Proctoring software integration, identity verification.

Requirement UFR-OE-110-04: Explaining Multiple Choice Answers

1. **Title:** Provide explanation for multiple choice answers.
2. **Actor:** Learner.
3. **Normal Flow:** Learner selects answer and optionally explains reasoning.

Requirement UFR-OE-110-05: Completing Essay Assignments

1. **Title:** Complete essay assignment.
2. **Actor:** Learner.
3. **Normal Flow:** Learner writes essay in text editor (supports formatting).

Requirement UFR-OE-110-06: Open Response Assessment Steps

1. **Title:** Follow ORA workflow.
2. **Actor:** Learner.
3. **Steps:** Submit response, self-assessment, peer assessment, receive score.

Requirement UFR-OE-110-07: How Grading Is Done In Open Response Assessments

1. **Title:** Understand ORA grading.
2. **Actor:** Learner.
3. **Description:** System displays grading rubric and peer review scores.

Requirement UFR-OE-110-08: Completing an Open Response Assessment

1. **Title:** Complete ORA.
2. **Actor:** Learner.
3. **Normal Flow:** Learner submits response, performs self/peer assessments.

Requirement UFR-OE-110-09: Receive Score and Provide Feedback

1. **Title:** Receive ORA score and provide feedback.
2. **Actor:** Learner.
3. **Normal Flow:** Learner views score and provides feedback on assessment experience.

Requirement UFR-OE-110-10: How Peer Assessment Scores Are Calculated

1. **Title:** Understand peer assessment scoring.
2. **Actor:** Learner.
3. **Description:** System explains scoring algorithm (average, median, adjusted).

Requirement UFR-OE-110-11: Canceled Responses

1. **Title:** Handle canceled responses.
2. **Actor:** Learner.
3. **Normal Flow:** Learner can cancel submission before assessment.

0.3.11 Team Projects Module

Requirement UFR-OE-111-01: About Teams and Topics

1. **Title:** Understand teams and topics.
2. **Actor:** Learner.
3. **Description:** System provides team and topic overview.

Requirement UFR-OE-111-02: Search for a Team

1. **Title:** Search for a team.
2. **Actor:** Learner.
3. **Normal Flow:** Learner searches teams by name, topic, or member.

Requirement UFR-OE-111-03: Join a Team

1. **Title:** Join an existing team.
2. **Actor:** Learner.
3. **Post:** Learner is added to team.

Requirement UFR-OE-111-04: Leave a Team

1. **Title:** Leave a team.
2. **Actor:** Learner.
3. **Post:** Learner is removed from team.

Requirement UFR-OE-111-05: Create a Team

1. **Title:** Create a new team.
2. **Actor:** Learner.
3. **Post:** Team is created; learner becomes team member.

Requirement UFR-OE-111-06: Participating in Team Discussions

1. **Title:** Participate in team discussions.
2. **Actor:** Learner.
3. **Normal Flow:** Team members discuss within team-specific forum.

0.3.12 Search Module**Requirement UFR-OE-112-01: Overview of Search**

1. **Title:** Understand search capabilities.
2. **Actor:** Learner.
3. **Description:** System supports full-text search across course content.

Requirement UFR-OE-112-02: Search a Single Course

1. **Title:** Search within a course.
2. **Actor:** Learner.
3. **Normal Flow:** Learner enters search query; system returns relevant content within course.

Requirement UFR-OE-112-03: Search All Your Courses

1. **Title:** Search across all courses.
2. **Actor:** Learner.
3. **Normal Flow:** Learner enters search query; system returns results from all enrolled courses.

0.3.13 **Bookmarking Module**

Requirement UFR-OE-113-01: Add or Remove a Bookmark

1. **Title:** Bookmark course content.
2. **Actor:** Learner.
3. **Normal Flow:** Learner clicks bookmark icon to save or remove bookmark.

Requirement UFR-OE-113-02: View Your Bookmarks

1. **Title:** View bookmarked content.
2. **Actor:** Learner.
3. **Normal Flow:** Learner accesses bookmarks page to view and navigate saved items.

0.3.14 **Notifications & Preferences Module**

Requirement UFR-OE-114-01: Where Notifications Appear

1. **Title:** Access notifications.
2. **Actor:** Learner.
3. **Normal Flow:** Notifications appear in dashboard, email, and in-app alerts.

Requirement UFR-OE-114-02: Activity Types

1. **Title:** Understand notification activity types.
2. **Actor:** Learner.
3. **Description:** System supports types: discussion replies, assignment due dates, course updates, etc.

Requirement UFR-OE-114-03: Managing Preferences

1. **Title:** Manage notification preferences.
2. **Actor:** Learner.
3. **Normal Flow:** Learner configures which notifications to receive and via which channels.

Requirement UFR-OE-114-04: Notifications Expiry

1. **Title:** Notification expiry.
2. **Actor:** System.
3. **Description:** Notifications expire after a defined period.

0.3.15 Course Wiki Module

Requirement UFR-OE-115-01: Using the Course Wiki

1. **Title:** Use the course wiki.
2. **Actor:** Learner.
3. **Normal Flow:** Learner accesses wiki from course navigation; views and edits (if allowed).

Requirement UFR-OE-115-02: View Live Course

1. **Title:** View live course content.
2. **Actor:** Learner.
3. **Normal Flow:** Learner accesses course content that is currently active/published.

0.4 Security and Compliance Requirements (Non-Functional)

This section defines mandatory security requirements derived from INSA's CMCSRS, the Computer Crime Proclamation, and the Electronic Signature Proclamation. All requirements are traceable to the *Security Architecture & Threat Model* document.

0.4.1 Access Control (CMCSRS 6.17)

- SEC-NFR-001-1.** The system must enforce role-based access control (RBAC) for all user actions. Roles include Learner, Educator, Administrator, and System.
- SEC-NFR-001-2.** All access to administrative functions must be restricted to users with appropriate privileges.
- SEC-NFR-001-3.** User accounts must be subject to a formal provisioning and de-provisioning process.
- SEC-NFR-001-4.** Default passwords for any system component must be changed upon first use.

SEC-NFR-001-5. The system must implement multi-factor authentication (MFA) for administrative accounts.

0.4.2 Cryptography (CMCSRS 6.18)

SEC-NFR-002-1. All data in transit must be encrypted using TLS 1.2 or higher.

SEC-NFR-002-2. All sensitive data at rest (passwords, API keys, PII) must be encrypted using strong, INSA-approved algorithms (e.g., AES-256).

SEC-NFR-002-3. Private keys and secrets must never be stored in plaintext; they must be managed via secure key management processes.

SEC-NFR-002-4. Digital signatures must be used for certificate verification as per the Electronic Signature Proclamation.

0.4.3 Network Security (CMCSRS 5.7.c)

SEC-NFR-003-1. A host-based firewall (UFW) must be active and configured to allow only necessary ports (22, 80, 443).

SEC-NFR-003-2. The internal network must be segmented into separate Docker networks for frontend, backend, and database services.

SEC-NFR-003-3. All unnecessary services and ports must be disabled.

SEC-NFR-003-4. Intrusion detection/prevention systems (fail2ban, IDS) must be deployed to protect against brute-force attacks.

0.4.4 Logging and Monitoring (CMCSRS 6.25)

SEC-NFR-004-1. All system and application logs must be collected centrally (SIEM) and retained for at least one month.

SEC-NFR-004-2. Security-relevant events (login failures, privilege changes, configuration changes) must be logged with timestamps and user identifiers.

SEC-NFR-004-3. Regular vulnerability scanning (e.g., Trivy) must be performed weekly on container images.

SEC-NFR-004-4. Anomaly detection and alerting must be configured for critical events.

0.4.5 Incident Response and Business Continuity (CMCSRS 6.26, 6.19)

SEC-NFR-005-1. An incident response plan must be documented and tested annually.

SEC-NFR-005-2. A business continuity plan with defined RTO and RPO must be maintained.

SEC-NFR-005-3. Regular backups of all critical data must be performed and stored off-site.

SEC-NFR-005-4. All security incidents must be reported to Ethio-CERT and INSA as per CMCSRS 6.26.f.

0.4.6 Personnel Security (CMCSRS 5.6, 6.37)

- SEC-NFR-006-1.** All personnel with access to sensitive systems must undergo security clearance.
- SEC-NFR-006-2.** Mandatory annual security awareness training must be provided to all employees.
- SEC-NFR-006-3.** Job descriptions must include security responsibilities.

0.4.7 Compliance and Audit (CMCSRS 6.21, 6.22, 6.23)

- SEC-NFR-007-1.** The system must undergo an annual security audit by an INSA-certified auditor.
- SEC-NFR-007-2.** All information systems must be accredited before deployment.
- SEC-NFR-007-3.** A compliance report must be submitted to INSA annually.

All security requirements are linked to the *Security Architecture & Threat Model* document, which provides detailed implementation guidance and risk assessment.

0.5 Traceability Matrix

The following table maps each requirement to its corresponding test case and security reference.

Requirement ID	Test Case ID	Security Reference
UFR-OE-001-01	TC-001	SEC-NFR-001, SEC-NFR-002
UFR-OE-001-03	TC-002	SEC-NFR-001
UFR-OE-003-01	TC-003	SEC-NFR-003, SEC-NFR-004
UFR-OE-003-02	TC-004	SEC-NFR-004
UFR-OE-004-01	TC-005	SEC-NFR-002
UFR-OE-004-03	TC-006	SEC-NFR-002
UFR-OE-005-03	TC-007	SEC-NFR-001, SEC-NFR-002
UFR-OE-006-03	TC-008	SEC-NFR-001, SEC-NFR-004
UFR-OE-103-01	TC-009	SEC-NFR-001
UFR-OE-106-03	TC-010	—
UFR-OE-107-01	TC-011	SEC-NFR-002
UFR-OE-109-01	TC-012	SEC-NFR-001, SEC-NFR-003
UFR-OE-110-02	TC-013	SEC-NFR-003
UFR-OE-112-02	TC-014	—
UFR-OE-113-01	TC-015	—
UFR-OE-114-03	TC-016	SEC-NFR-006
Continued		

Requirement ID	Test Case ID	Security Reference
SEC-NFR-001 to 007	See Security document	CMCSRS, Proc.958/2016

0.6 References

- *Security Architecture & Threat Model – Moove Education Platform* (Security_OpenEdX.tex)
- CMCSRS (Critical Mass Cyber Security Requirement Standard) v2.0 (2025)
- FDRE National Cybersecurity Policy (2024)
- Computer Crime Proclamation No. 958/2016
- Electronic Signature Proclamation No. 1072/2018

Apache License 2.0 Attribution

This product includes software developed by the **Open edX Project** (<https://open.edx.org/>) licensed under the Apache License, Version 2.0 (<http://www.apache.org/licenses/LICENSE-2.0>).

Moove Education Platform is a derivative work of Open edX. No modifications have been made to the original license terms. A copy of the full license text is available in the root of the source repository and in Appendix A of this document.

.1 Full Apache License 2.0 Text